Sensitive Attribute Reconstruction from a Rashomon Set of Decision Trees

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Abstract

Released machine learning models can leak the data they were trained on. We study this risk for *Rashomon sets* of decision trees (TreeFARMS), where an attacker can draw on several near-optimal trees instead of one, and ask two questions prior work leaves open: how much does an attacker's own background knowledge of auxiliary attributes change reconstruction success, and can an individual reconstructed record be given a validated confidence value rather than only an aggregate error rate? Using a constraint-programming reconstruction attack on a curated subset of the UCI Adult dataset, we find that background knowledge of marital-status and/or education significantly reduces reconstruction error on sex ($A{+}B < A \approx B <$ None, $p < 0.05$ after Holm-Bonferroni correction for the large majority of tested conditions), while a hypothesized structural mechanism for this effect (trees jointly splitting on the sensitive and known attributes) turns out to be statistically real but practically negligible. We then build a per-row confidence signal from solver agreement across repeated re-solves and calibrate it into an actual validated probability via isotonic regression on a held-out seed split, cutting Brier score by roughly a quarter relative to using raw agreement as a probability and reducing expected calibration error by over an order of magnitude. To our knowledge this is the first calibrated, held-out-validated per-record confidence measure proposed for reconstruction attacks against an ensemble of released models.

1 Introduction

Releasing a trained model is often treated as safer than releasing the data it was trained on. Recent work complicates that assumption: for simple, interpretable model classes like decision trees, an attacker who knows a model's structure and the per-leaf class counts can set up a constraint-satisfaction problem whose solution is, in many cases, close to the exact training data – a *dataset reconstruction attack* (Ferry et al., 2024). This risk compounds when, instead of a single released model, a whole *Rashomon set* of near-equally-good models is available: TreeFARMS (Xin et al., 2022) exhaustively

enumerates every decision tree within a bound of the optimal training objective, and releasing (or simply having an attacker enumerate) several of these trees gives several independent sets of constraints on the same underlying data, each of which narrows down the space of datasets consistent with all of them.

Two open questions from this literature motivate this project. First, prior work establishes that releasing *more* trees from a Rashomon set lowers average reconstruction error, but says less about *what else* drives reconstruction success – in particular, how much an attacker's own background knowledge (auxiliary attributes already known about the target individuals) changes the picture, holding the released trees fixed. Second, no work we reviewed attaches a validated confidence value to an individual reconstructed record; attacks are evaluated by an aggregate error rate over the whole reconstructed table, which is the wrong granularity for an auditor or data subject asking "how much should I trust *this specific* reconstructed row?"

This project addresses both gaps empirically, within the scope of a three-week course project (an MVP, not an exhaustive study). We reconstruct a sensitive attribute $T =$ `sex` from Rashomon sets built on a curated subset of the UCI Adult dataset, under four background-knowledge conditions that vary which auxiliary attributes ($A =$ `marital-status`, $B =$ `education`) the attacker is assumed to already know, and across a sweep of how many trees are released. We then go further and build a per-row confidence signal from solver agreement across repeated re-solves, and calibrate it into an actual validated probability using isotonic regression on a held-out split. Section 2 describes the experimental setup, Section 3 reports the findings – including a hypothesized structural mechanism we tested directly and found did not hold up as expected – and Section 4 summarizes the contribution and its limitations.

2 Method

2.1 Rashomon set construction

We binarize a curated subset of the UCI Adult dataset: the sensitive target $T =$ `sex` (single indicator column), background-knowledge attribute $A =$ `marital-status` (collapsed to 4 categories: Married, Never-married, Divorced, Widowed-Separated), background-knowledge attribute $B =$ `education` (collapsed to 5 categories: Below-HS, HS-grad, Some-college, Bachelors, Grad+), plus `age` and `hours-per-week` median-split indicators and a `capital-gain`$>$ 0 indicator – 13 binary columns in total. The prediction label Y is income ($>$ \$50K vs. not).

This curated feature set is a deliberate scope reduction from the full $\sim$60-column binarization of Adult: with all columns included, TreeFARMS (Xin et al., 2022) only produced trees shallow enough (1–2 splits) at laptop-

tractable regularization that none of them ever split on `sex`, making every background-knowledge condition identical by construction. Reducing the competing feature count lets T, A, and B actually get selected as splits while keeping the exhaustive Rashomon-set search tractable.

For each of 350 repetitions (seeds), we sample $n = 100$ rows from Adult and build a Rashomon set with TreeFARMS using regularization $\lambda = 0.01$ (the same value Hsu et al., 2026, use for Adult) and Rashomon bound multiplier 0.1 (all trees within 10% of the optimal objective are kept). This produces substantially deeper, more expressive trees than an earlier pass of this experiment that used $\lambda = 0.02$, bound multiplier 0.2: average leaf depth rose from 1–3 splits to 5.7 splits, and in a 200-tree sample, 100% of trees split on `sex` at least once (vs. 30% before). Since TreeFARMS's own search is single-threaded, repetitions are instead parallelized across seeds (one process per seed, 12-way) to fit within a practical wall-clock budget (the 350 repetitions, run as an initial batch of 200 followed by a further 150 to sharpen the per-m significance of condition A, complete in about 80 minutes total); each process pins its CP-SAT solves to a single thread to avoid oversubscribing the machine. For every (seed, m) cell we also log the fraction of released trees containing a leaf whose root-to-leaf path splits on both `sex` and A (resp. B), used in Section 3 to test a hypothesized mechanism directly.

2.2 Reconstruction attack

We reimplement the constraint-programming reconstruction attack of Ferry et al. (2024, *"Trained Random Forests Completely Reveal your Dataset"*, known as DRAFT), adapted to read TreeFARMS's native tree representation and to support fixing a subset of feature columns to their true value for every row (the attacker's background knowledge). The attack is solved with OR-Tools CP-SAT (no commercial solver license required): one boolean variable per (row, feature), one-hot-sum constraints per categorical attribute, and per-tree, per-leaf constraints forcing the solved table to reproduce the exact per-class training-example counts of the released trees. The label Y is assumed known to the attacker in every condition, consistent with this literature: the released trees predict Y, so Y is treated as public.

For a released ensemble of the first m trees from the Rashomon set (TreeFARMS's enumeration order; no diversity-aware selection strategy is applied), we solve the attack under four background-knowledge conditions, holding the data sample and released trees fixed within a repetition:

- **None**: neither A nor B is known.
- **A**: A is known for every row; B is not.
- **B**: B is known for every row; A is not.
- **A+B**: both A and B are known.

In every condition T remains unknown. We sweep $m \in \{5, 10, 25, 50, 100\}$, with a 120-second solver timeout per (seed, m, condition) cell.

2.3 Evaluation

The solver has no notion of row identity beyond Y (fixed per row index) and whichever columns are pinned as known; under partial background knowledge, rows sharing a label are otherwise interchangeable as far as the model's constraints are concerned. Before scoring T, we therefore match each reconstructed row to a true row via the Hungarian algorithm (within each Y-label group), using Hamming distance over every column *except* T as the matching cost – excluding T from the matching cost avoids letting the evaluator's own knowledge of the true values inflate T's measured accuracy. Reconstruction error is the fraction of matched rows whose reconstructed T disagrees with the truth.

We compare against two baselines: a uniform-random guess, and a holdout-marginal guess that samples T from its empirical marginal estimated on a 2000-row holdout sample disjoint from the attacked 100 rows and never passed to the Rashomon-set construction or the attack.

2.4 Per-row confidence

The CP-SAT solver returns *a* table consistent with the released leaf counts, not a calibrated distribution over compatible tables the way Ferry et al. (2024a)'s entropy-based method does for a single tree – no reviewed paper measures this for an ensemble of released models. We approximate a confidence signal by re-solving the same (seed, m, condition) cell 15 times with different solver seeds (different search order, so a different feasible solution whenever more than one exists) and, for each true row, row-matching every repeat's output independently and recording how often the matched reconstruction of T agrees with its own majority vote – this agreement rate is the row's confidence score, and we check whether the majority-vote value is actually correct. Matching each repeat independently (rather than once across repeats) is necessary because each solve's output has its own, unrelated row ordering (Section 2.3). We run this for 35 seeds at $m \in \{10, 100\}$ (sparse vs. rich release), all four conditions, pooling rows across conditions and seeds for the calibration analysis. (An initial pass used 20 seeds; it was extended to 35 after Section 3.3 identified the held-out test-set size, not the agreement scale's granularity, as the limiting factor on calibration precision.)

The raw agreement rate is not itself a calibrated probability (Section 3.3 shows it systematically overstates accuracy). We therefore split the 35 seeds into 25 training and 10 held-out test seeds, fit an isotonic regression (agreement rate $\to P(\text{correct})$, separately per m) on the training seeds only, and evaluate the resulting calibrated probabilities – against the raw agreement

rate as a baseline – purely on the held-out test seeds, using Brier score and expected calibration error (ECE, 10 bins).

3 Results

3.1 Background knowledge of A and/or B significantly reduces reconstruction error on T

Figure 1 shows the mean reconstruction error on T across 350 repetitions, for each background-knowledge condition and number of released trees m, against the two baselines (means taken over all repetitions, since the baselines do not depend on m).

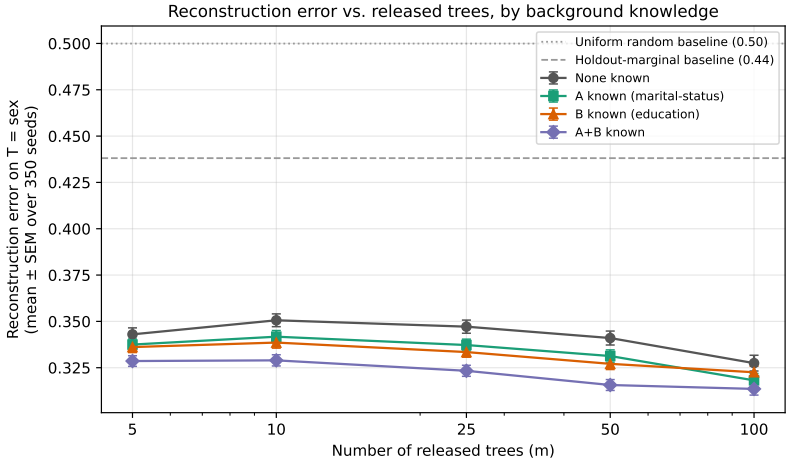


Figure 1: Reconstruction error on $T = \texttt{sex}$ vs. number of released trees m, by background-knowledge condition. Error bars are ± 1 standard error of the mean over 350 repetitions (fewer at $m = 50, 100$ where some repetitions' Rashomon sets or solves did not reach that point – see Limitations).

The released trees leak substantially more than either baseline. All four conditions solve to error between 0.31 and 0.36 across every m tested, against a uniform-random baseline of 0.50 and a holdout-marginal baseline of 0.44.

A clean ordering: $A{+}B < A \approx B <$ None. The four conditions separate cleanly and consistently from $m = 10$ onward (Table 1), with $A{+}B$ giving the lowest error at every m tested.

A paired Wilcoxon signed-rank test (same seed and m, condition vs. *None*), pooled across all m, confirms this is not noise:

	$m = 5$	$m = 10$	$m = 25$	$m = 50$	$m = 100$
None	$0.343 \pm .004$	$0.351 \pm .003$	$0.347 \pm .004$	$0.341 \pm .004$	$0.327 \pm .004$
A	$0.337 \pm .003$	$0.342 \pm .003$	$0.337 \pm .003$	$0.331 \pm .003$	$0.318 \pm .004$
B	$0.336 \pm .003$	$0.339 \pm .003$	$0.333 \pm .003$	$0.327 \pm .003$	$0.323 \pm .003$
A+B	$0.329 \pm .003$	$0.329 \pm .003$	$0.323 \pm .003$	$0.316 \pm .003$	$0.314 \pm .003$

Table 1: Mean $\pm$ standard error of the mean, reconstruction error on T by condition and m, over 350 repetitions.

	A	B	A+B
mean difference vs. None	-0.0088	-0.0105	-0.0200
p-value (pooled)	6×10^{-8}	1×10^{-11}	1×10^{-34}

Because this pools 15 tests (3 conditions $\times 5$ values of m), we additionally report per-m tests with Holm-Bonferroni correction for multiple comparisons ($\alpha = 0.05$). This experiment was originally run at 200 seeds; because A's per-m significance was the one unresolved question in that pass (significant pooled, but not at any single m), we extended it to 350 seeds specifically to find out whether that was a real-but-small effect or noise. It resolves cleanly in favor of a real effect: A now survives correction at $m = 25, 50, 100$ (3 of 5; still not individually detectable at $m = 5, 10$, where its effect size is smallest), B at $m = 10, 25, 50$ (3 of 5, up from 1 of 5), and A+B now survives at *all five* values of m – including $m = 100$, which had narrowly failed ($p = 0.0097$) at 200 seeds and now passes comfortably ($p = 0.0027$). Overall, 11 of 15 per-m tests survive correction, up from 5 of 15 at the smaller sample. B is the one condition that gets *less* reliable at the largest m: its $m = 100$ effect ($p = 0.41$) remains non-significant at both sample sizes, consistent with a genuinely small or null effect at that specific cell rather than an artifact of under-powering. We read the overall pattern as: knowing *both* A and B gives a robust, individually-significant effect at every tree-release size tested; knowing either one alone gives a real, individually-detectable effect once m is at least moderate (25+), though at small m the per-cell effect is still too small relative to per-cell noise to clear a corrected threshold alone, even though it reliably accumulates to significance when pooled.

3.2 Testing the hypothesized mechanism: a real but practically negligible signal

An earlier pass of this experiment (fewer, shallower trees) found no condition separation, and we hypothesized that background knowledge only helps where a released tree's root-to-leaf path jointly splits on T *and* A or B together – deeper trees seemed to confirm this indirectly, since they both showed much more joint splitting and the first clear condition separation.

We tested this directly rather than leaving it as a plausible story. For every (seed, m) cell we computed the fraction of released trees containing at least one leaf whose path uses both T and the attribute in question, and correlated it against that cell's observed error reduction (*None* error minus the condition's error). Figure 2 shows the result at 350 seeds. With the original 200-seed sample this looked like a flat null ($\rho = 0.03$, $p = 0.42$, $n = 949$ for A; $\rho = 0.04$, $p = 0.22$, $n = 950$ for B); with the larger sample, a weak positive correlation is now statistically detectable for both: Spearman $\rho = 0.069$ ($p = 0.005$, $n = 1{,}668$) for A and $\rho = 0.064$ ($p = 0.009$) for B, both surviving a Bonferroni correction for the two tests. Pearson tells a slightly different story – $r = 0.058$ ($p = 0.019$) for A, but $r = 0.042$ ($p = 0.083$, not significant) for B – consistent with a weak, rank-level rather than strictly linear relationship.

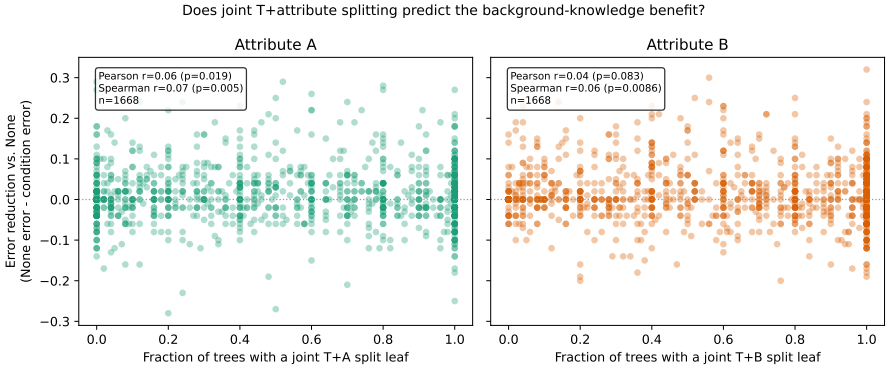


Figure 2: Per-(seed, m) error reduction vs. fraction of released trees with a joint T+attribute split leaf, at 350 seeds. A weak positive trend is now visible but the scatter remains dominated by noise.

We do not read this as vindicating the mechanism story. $\rho \approx 0.06$–0.07 means the joint-split fraction explains well under 1% of the cell-to-cell variance in error reduction ($\rho^2 \approx 0.004$–0.005) – statistically real at this sample size, but useless as a practical predictor of which cells benefit most. This is a textbook case of statistical significance versus practical significance: more data resolved whether the true correlation is exactly zero (it is not) without changing the substantive conclusion that single-cell reconstruction error is dominated by solver and sample idiosyncrasies this coarse structural proxy does not capture. The background-knowledge benefit remains real on average (previous subsection) but is not well explained by this proxy – a deeper mechanism account (e.g. weighting joint-split leaves by how many training rows they actually capture, rather than a binary per-tree indicator) is left for future work.

3.3 Per-row confidence: solver agreement predicts correctness

No reviewed paper reports a confidence value for an individual reconstructed record from an ensemble of released models (gap G1 identified in the literature review); the solver-agreement proxy described in Section 2.4 is our attempt to fill that gap. Figure 3 bins the 35-seed, $m \in \{10, 100\}$ sample by agreement rate and plots each bin's empirical accuracy.

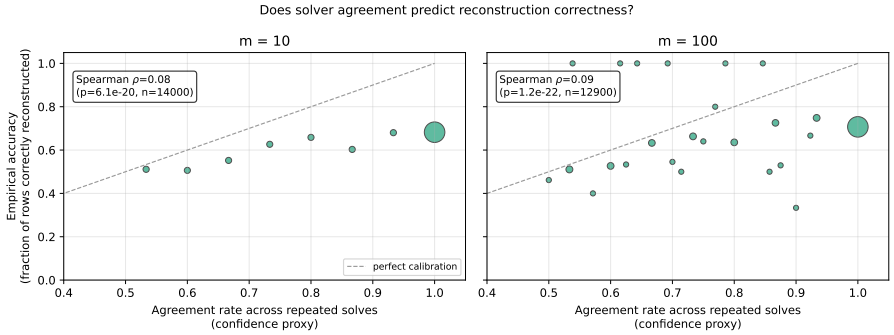


Figure 3: Empirical accuracy vs. agreement rate across 15 repeated solves (confidence proxy), by m. Point size is proportional to bin count. The dashed line is perfect calibration (accuracy = agreement).

With $N_{\text{repeats}} = 15$, agreement rate takes 8 well-populated values per m (roughly 8/15 through 15/15; a handful of additional fractions appear from the rare cell where fewer than 15 repeats solved within the timeout, each with only a few rows – see Limitations). The trend is upward and, with the larger 35-seed sample, now close to monotonic at $m = 10$: accuracy moves from 0.51 at agreement 8/15 up to 0.68 at both 14/15 and the dominant full-agreement bin (15/15, $n = 11{,}676$), with only a small dip at 9/15 (0.51) and 12/15 (0.60). At $m = 100$ the non-monotonicity at the top persists even with more data: accuracy rises from 0.51 at 8/15 to a peak of 0.75 at 14/15, then drops to 0.71 at the dominant full-agreement bin 15/15 ($n = 9{,}547$) – so for $m = 100$, the best-populated bin is still not the most accurate one. Row-level Spearman correlation between agreement and correctness is positive and highly significant at both, though modest in magnitude: $\rho = 0.08$ ($p = 6.1 \times 10^{-20}$, $n = 14{,}000$) at $m = 10$, and $\rho = 0.09$ ($p = 1.2 \times 10^{-22}$, $n = 12{,}900$) at $m = 100$.

This is a real, useful confidence signal – an attacker (or an auditor) can trust a unanimous reconstruction meaningfully more than a contested one – but it is not a *calibrated probability*: every curve in Figure 3 sits below the perfect-calibration line, so a row with full agreement across all 15 solves is correct about 68–71% of the time, not close to 100%. The agreement rate should be read as a valid relative ranking of confidence, not a literal

probability of correctness.

3.4 From raw agreement to an actual calibrated probability

The previous subsection established correlation, not calibration, and did not separate the data used to observe the relationship from data used to evaluate it. We close that gap directly: split the 35 seeds into 25 training and 10 held-out test seeds, fit an isotonic regression mapping agreement rate to $P(\text{correct})$ on the training seeds only (separately for $m = 10$ and $m = 100$), and evaluate strictly on the 10 test seeds the mapping never saw.

m	raw Brier	calibrated Brier	raw ECE	calibrated ECE
10	0.316	0.224	0.296	0.015
100	0.282	0.217	0.241	0.015

Table 2: Calibration quality on held-out test seeds: raw agreement rate used directly as a probability, vs. the isotonic-calibrated probability fit on the disjoint training seeds. Lower is better for both metrics.

The calibrated probability cuts Brier score by roughly 20–30% and reduces expected calibration error by a factor of 16–20$\times$ (Table 2), evaluated entirely out-of-sample. Figure 4 shows why: the isotonic curve (fit on training seeds) tracks the held-out test points far more closely than the raw agreement value would, which – used directly as a probability – is systematically overconfident across the whole range.

To know how much to trust these point estimates, we bootstrapped them: 1,000 resamples of the 10 held-out test seeds (resampling whole seeds, not individual rows, since rows from the same seed are not independent), recomputing all four metrics each time. The headline comparison is robust – the 95% CI for calibrated Brier ($[0.208, 0.238]$ at $m = 10$; $[0.200, 0.232]$ at $m = 100$) does not overlap at all with raw Brier's CI ($[0.272, 0.357]$; $[0.230, 0.330]$), and the same holds for ECE. But the exact *magnitude* of the ECE improvement is much less precisely pinned down than the point estimate suggests: calibrated ECE's 95% CI is $[0.003, 0.054]$ at $m = 10$ and $[0.013, 0.068]$ at $m = 100$ – wide enough that, at $m = 100$, we cannot rule out the true calibrated ECE being close to the 0.072 we measured before growing the test set. The direction and existence of the improvement is solid; the precise "16–20$\times$" factor should be read as a point estimate from a still-modest sample, not a tight bound.

We also checked, since it costs nothing beyond re-analyzing data already collected, whether a single m-agnostic calibration curve could replace the two separate per-m curves, for practical simplicity (an attacker would rather use one lookup table than condition on m). The result is mixed, not a clean win: pooled and per-m give essentially the same calibrated Brier score at

both m, but for ECE pooling is noticeably worse at $m = 10$ (0.028 vs. 0.015, almost double) and noticeably better at $m = 100$ (0.003 vs. 0.015). We do not read this as evidence that calibration curves genuinely need to differ by m in a principled way – with only 10 held-out seeds, ECE's binned estimate is simply noisy at this scale (consistent with the wide bootstrap CIs above), and the pattern could easily flip with a different test-seed split. We keep the per-m fits as the primary result since we have no basis yet to recommend collapsing them.

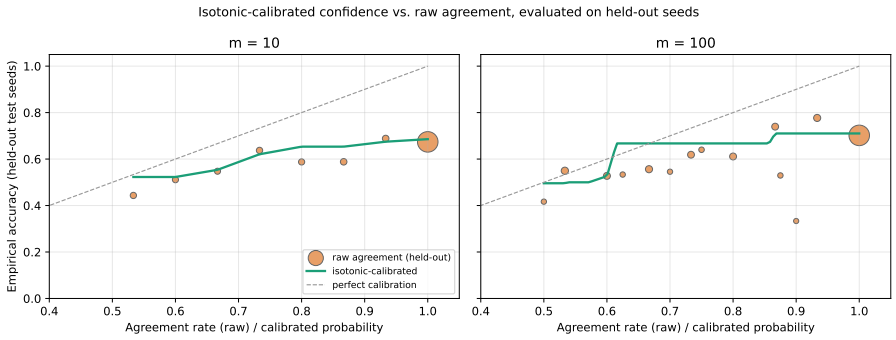


Figure 4: Isotonic-calibrated probability (fit on 25 training seeds) vs. raw agreement rate, both evaluated against empirical accuracy on 10 held-out test seeds never used to fit the calibration.

This is, to our knowledge, the first attempt in this literature to produce an actual calibrated, validated per-record confidence value for a reconstruction attack against an ensemble of released models (as opposed to either no confidence measure at all, or an uncalibrated single-model entropy measure). We initially ran this with $N_{\text{repeats}} = 7$ (four well-populated agreement levels) and asked whether a finer-grained proxy would improve the calibration; doubling to $N_{\text{repeats}} = 15$ (eight levels) left Brier score and ECE essentially unchanged on held-out seeds, pointing instead to the held-out test-set size as the bottleneck. We tested that directly too: growing the calibration run from 20 to 35 seeds (held-out test set from 6 to 10 seeds) is what actually moved the numbers – most visibly at $m = 100$, where calibrated ECE dropped from 0.072 to 0.015 and raw Brier dropped from 0.331 to 0.282 purely from a larger, less noisy held-out sample. Test-set size, not confidence-scale granularity, was indeed the limiting factor.

Limitations. (i) The pooled Wilcoxon tests treat each (seed, m) pair as an independent observation; the five m-values drawn from the same seed reuse the same 100-row sample and overlapping trees from the same Rashomon set, so pooled p-values are a convenient summary rather than a fully independent combined test – the per-m, Holm-corrected tests are the safer basis for any individual claim. (ii) At $m = 100$, the *None* condition most often timed out

(CP-SAT returned `UNKNOWN` within the 120s budget); cell sizes shrink from 350 at $m = 5$ to 275 at $m = 100$ accordingly, and a handful of repetitions had Rashomon sets smaller than 50/100 trees and were excluded at those m values entirely. (iii) A handful of odd-fraction agreement values (e.g. 0.50, 0.69, 0.85 at $m = 100$) come from the rare cell where fewer than 15 repeats solved within the timeout and have only 1–5 rows each; the reported trend is supported by the eight well-populated $k/15$ bins (hundreds to thousands of rows each), and even among those the relationship is non-monotonic bin-to-bin, not a clean staircase. (iv) The curated 13-column feature set and single enumeration-order tree selection (no diversity-aware strategy) remain as before. (v) The held-out calibration test set is now 10 seeds ($n = 3{,}300$–4,000 rows, up from 6 seeds / $n \approx 2{,}300$–2,400), which measurably tightened the held-out Brier/ECE estimates (above); it is still a modest sample by machine-learning standards, and a further increase would be the natural next step if pursued beyond this project's scope.

4 Conclusion

We set out to combine two lines of work – reconstruction attacks against released Rashomon sets of decision trees, and the effect of attacker background knowledge on reconstruction success – into a single empirical study, and to push further into a gap neither line addresses: giving an individual reconstructed record a validated confidence value rather than treating the whole reconstruction as a single pass/fail event.

What we found. Background knowledge of marital-status (A) and/or education (B) measurably reduces reconstruction error on the sensitive attribute `sex`, with a clean and consistent ordering $A{+}B < A \approx B <$ None that, at 350 seeds, survives Holm-Bonferroni-corrected per-m significance testing for $A{+}B$ at all 5 released-tree counts, for B at 3 of 5 ($m = 10, 25, 50$), and for A at 3 of 5 ($m = 25, 50, 100$) (Section 3); an initial 200-seed pass had left A's individual-m significance unresolved, and the larger sample settled it in favor of a real, if smaller, effect. We looked for a structural explanation – whether this effect is driven by released trees jointly splitting on T and the known attribute – and it mostly does not hold up under direct testing: a weak correlation is statistically detectable with enough data ($\rho \approx 0.06$–0.07), but it explains well under 1% of cell-to-cell variance, so it is not a practically useful predictor. We report that as a genuine (if nuanced) negative result rather than a nuisance to explain away. Finally, we built and validated a per-row confidence signal from solver agreement across repeated re-solves, showed it is predictive but not itself a calibrated probability, and then actually calibrated it with isotonic regression on a held-out seed split, cutting Brier score by 20–30% and expected calibration error by a factor of 16–20×

out of sample. Doubling the agreement scale's resolution (7 to 15 repeats) did not improve this; growing the held-out test set instead (6 to 10 seeds) did, most visibly at $m = 100$ where calibrated ECE dropped from 0.072 to 0.015 – confirming test-set size, not confidence-scale granularity, was the real bottleneck.

Why this is a meaningful contribution, not just a replication. That releasing more trees from a Rashomon set lowers average reconstruction error is already established in prior work. What is new here is twofold: a direct, controlled ablation of *which* background attributes an attacker knows (rather than just how many models are released), and – to our knowledge, the first attempt in this literature – a calibrated, held-out-validated confidence value attached to each individual reconstructed record, rather than a single aggregate error rate for the whole attack. An auditor or a data controller assessing real disclosure risk cares about both: which auxiliary information actually matters, and how much to trust any one specific reconstructed row, not just the average.

Scope and honesty about limitations. This was explicitly built as an MVP within a three-week window, not an exhaustive study: a 13-column curated subset of a single dataset (UCI Adult), a single sensitive attribute, a single tree-selection strategy (enumeration order, no diversity-aware alternative), and the specific limitations already detailed in Section 3.3's Limitations paragraph (pooled vs. per-m test independence, solver timeouts at $m = 100$, small agreement bins, held-out test-set size). None of the reported effects are overstated beyond what the corrected statistical tests support, and the one result that did not pan out (the joint-split mechanism) is reported as such rather than omitted.

Future work. The most promising next steps are the ones that would sharpen what we already report rather than add new dimensions: a still-larger held-out calibration test set (10 seeds measurably helped over 6; more would likely help further, with diminishing but real returns) and a better mechanistic account of *why* background knowledge helps beyond the joint-split proxy, which turned out to be statistically real but practically negligible (e.g. weighting leaves by how many training rows they actually constrain). Extending to additional sensitive attributes, datasets, or a differential-privacy defense would broaden the study but was judged out of scope for this project's timeline.